

8.1 Product/Quotients of Rational Expressions

Date _____ Period _____

Simplify each expression.

1)
$$\frac{(n+17)(n-19)}{n+13} \cdot \frac{n+13}{(n-19)(-n+19)}$$

2)
$$\frac{1}{56r^2} \cdot \frac{17r(r+19)}{r+19}$$

3)
$$\frac{7-x}{(x+3)(x-7)} \cdot \frac{(x+17)(x+3)}{x-14}$$

4)
$$\frac{(x-3)(x+2)}{x-3} \cdot \frac{x-5}{2x(x+2)}$$

5)
$$\frac{16v(v-8)}{16v} \cdot \frac{11(v-10)}{(10-v)(v-8)}$$

6)
$$\frac{x+13}{14x} \div \frac{20(4-x)}{20(x-4)}$$

7)
$$\frac{10(k-15)}{5(15-k)} \div \frac{10(k+9)}{k+6}$$

8)
$$\frac{x-3}{x+16} \div \frac{(x-11)(x-3)}{4x^2(x+16)}$$

9)
$$\frac{16(m-5)}{m-5} \div \frac{7(m+11)}{7}$$

10)
$$\frac{3}{n-8} \div \frac{3(n-8)}{8(n-8)}$$

$$11) \frac{2v}{v^2 - v - 42} \div \frac{1}{v + 6}$$

$$12) \frac{1}{x - 2} \cdot \frac{8x + 48}{x + 6}$$

$$13) \frac{7}{p - 1} \div \frac{21p + 14}{6p^2 + 4p}$$

$$14) \frac{4n - 28}{4n} \cdot \frac{4n}{n - 7}$$

$$15) \frac{1}{n - 5} \div \frac{n + 5}{n^2 + 2n - 35}$$

$$16) \frac{n^2 - 20n + 100}{8n^3 + 64n^2} \cdot \frac{n^2 + 16n + 64}{n - 10}$$

$$17) \frac{n + 5}{n^2 + n - 2} \div \frac{5n + 25}{5n + 10}$$

$$18) \frac{3x - 27}{4x^2 - 36x} \cdot \frac{4x^2 + 36x}{6x^2 + 54x}$$

$$19) \frac{n^2 - 14n + 49}{2n - 14} \cdot \frac{2n - 14}{n - 7}$$

$$20) \frac{v + 5}{6v^3 + 30v^2} \cdot \frac{v^2 + 3v - 28}{10v - 40}$$

Answers to 8.1 Product/Quotients of Rational Expressions (ID: 1)

1) $\frac{n+17}{-n+19}$

2) $\frac{17}{56r}$

3) $\frac{-x-17}{x-14}$

4) $\frac{x-5}{2x}$

5) -11

6) $\frac{-x-13}{14x}$

7) $\frac{-k-6}{5(k+9)}$

8) $\frac{4x^2}{x-11}$

9) $\frac{16}{m+11}$

10) $\frac{8}{n-8}$

11) $\frac{2v}{v-7}$

12) $\frac{8}{x-2}$

13) $\frac{2p}{p-1}$

14) 4

15) $\frac{n+7}{n+5}$

16) $\frac{(n-10)(n+8)}{8n^2}; \{0, -8, 10\}$

17) $\frac{1}{n-1}; \{1, -2, -5\}$

18) $\frac{1}{2x}; \{0, 9, -9\}$

19) $n-7; \{7\}$

20) $\frac{v+7}{60v^2}; \{0, -5, 4\}$